CHECKLIST DATA PRE-PROCESSING

BASIC CHECKLIST

Data Collection, Inspection and Cleaning

1) ☐ Verify data sources and ensure data integrity.
2) ☐ Examine data distribution and variable types.
3) ☐ **Check for Inconsistencies.**
   a) ☐ **Format Inconsistencies**
      i) Inconsistent formatting of categorical variables (e.g., "Male" vs. "male").
      ii) Inconsistent date or time formats (e.g., "MM/DD/YYYY" vs. "YYYY-MM-DD").
      iii) Mixed case or variations in spelling (e.g., "Very Good" vs. "V.Good").
   b) ☐ **Duplicate Records**
      i) Presence of duplicate rows or observations in the dataset.
      ii) Duplicate entries for unique identifiers such as customer IDs or transaction IDs.
   c) ☐ **Logical Inconsistencies**
      i) Values that violate logical constraints (e.g., negative values for age or income).
      ii) Inconsistent relationships between variables (e.g., a person's age conflicting with their reported education level).
   d) ☐ **Data Entry Errors**
      i) Typos or misspellings in categorical variables.
      ii) Outliers or extreme values that are likely due to data entry mistakes.
4) ☐ **Check for Missing Values.**
   a) ☐ **Identifying Missing Values**
      i) Look for cells or entries in your dataset that contain blanks, nulls, or placeholders like "NA" or "NaN".
      ii) Use summary statistics or visualizations to identify patterns of missingness across variables.
      iii) Check for missing values using functions like isnull() in Python.
   b) ☐ **Handling Missing Values**
      i) Decide on an appropriate strategy for handling missing values based on the percentage of missingness and the nature of your data:
      ii) If the percentage of missing values is small (e.g., less than 5-10%), consider imputation techniques such as mean, median, mode, or predictive imputation.
      iii) If the percentage of missing values is large (e.g., more than 10-20%), evaluate whether dropping rows or columns with missing values is feasible and won't result in significant loss of information.

iv) If missing values are systematically related to other variables, consider using more sophisticated imputation methods or modeling approaches to predict missing values.
v) Document the approach taken to handle missing values and any assumptions made during the process.

c) ☐ Imputation Techniques

i) **Mean/Median/Mode Imputation**: Replace missing values with the mean, median, or mode of the respective variable.

ii) **Predictive Imputation**: Use machine learning algorithms to predict missing values based on other variables in the dataset.

iii) **Multiple Imputation**: Generate multiple imputed datasets to account for uncertainty in imputed values.

iv) **Other Advanced Techniques**: Consider methods like hot-deck imputation, K-nearest neighbors (KNN) imputation, or interpolation for handling missing values.

d) ☐ **Data Quality Considerations**

i) Evaluate the impact of missing values on the analysis or modeling task.
ii) Consider potential biases introduced by handling missing values, especially if missingness is not completely at random (MCAR) but related to the outcome or other variables in the dataset.
iii) Perform sensitivity analyses to assess the robustness of results to different imputation strategies or handling methods.

5) ☐ <u>Check for Outliers</u>

a) ☐ Identifying Outliers

i) Use graphical methods such as box plots, histograms, or scatter plots to visually inspect the distribution of each variable and identify potential outliers.
ii) Calculate summary statistics like mean, median, standard deviation, and quartiles to quantitatively assess the spread of the data and identify observations that fall far from the central tendency.
iii) Consider domain knowledge and context-specific thresholds to determine what constitutes an outlier in your dataset.

b) ☐ Handling Outliers

i) Depending on the nature of the outliers and the analysis or modeling task, you may choose one or more of the following approaches.

(1) **Removal:** Delete outliers from the dataset if they are deemed to be data entry errors or if they significantly skew the distribution of the data. However, be

cautious about removing outliers without a valid reason, as this could lead to loss of valuable information.

(2) **Transformation:** Apply mathematical transformations such as log transformation, square root transformation, or Box-Cox transformation to make the data more normally distributed and reduce the impact of outliers.

(3) **Winsorization:** Winsorize the data by replacing extreme values with less extreme values (e.g., replacing outliers with the 95th or 99th percentile value).

(4) **Capping:** Cap extreme values by setting a threshold beyond which values are truncated or replaced with a specified value.

(5) **Model-Based Approaches:** Use robust statistical methods or machine learning algorithms that are less sensitive to outliers, or incorporate outlier detection techniques into the modeling process.

6) ☐ **Feature Engineering**

a) ☐ Create new features based on domain knowledge or feature extraction techniques.

b) ☐ Encode categorical variables.

i) One-hot encoding, Label encoding, or Target encoding.

c) ☐ Handle temporal variables appropriately

i) Extract relevant features, Create time-based features, etc.

ii) Consider feature scaling for algorithms sensitive to feature magnitudes.

7) ☐ **Data Transformation**

a) ☐ Apply transformations for skewed distributions: log transformation, Box-Cox transformation, etc.

b) ☐ Handle multicollinearity: remove highly correlated features or apply dimensionality reduction techniques like PCA.

c) ☐ Deal with non-linear relationships: consider polynomial features or nonlinear transformations.

ADVANCE CHECKLIST

1) ☐ **Data Splitting**
   a) Split the dataset into training, validation and test sets to evaluate model performance.
   b) Ensure stratified sampling if dealing with imbalanced classes.
   c) Randomize data splitting to avoid any biases in the order of samples.

2) ☐ **Data Normalization**
   a) Normalize numerical features if necessary, ensuring they have zero mean and unit variance.
   b) Apply min-max scaling to bound features within a specific range if needed.

3) ☐ **Handling Imbalanced Data**
   a) Implement techniques like oversampling, undersampling, or synthetic data generation to address class imbalance.
   b) Utilize appropriate evaluation metrics (e.g., F1-score, ROC-AUC) for imbalanced datasets.

4) ☐ **Feature Selection**
   a) Perform feature selection techniques such as filter methods, wrapper methods, or embedded methods to identify the most relevant features.
   b) Conduct feature importance analysis using algorithms like decision trees or feature importance scores from ensemble methods.

5) ☐ **Addressing Data Leakage**
   a) Ensure no leakage of information from the validation or test set into the training set.
   b) Remove any features derived from the target variable or that contain future information.

6) ☐ **Documentation and Version Control**
   a) Document all steps taken during data preprocessing for reproducibility.
   b) Version control the dataset and preprocessing scripts to track changes and facilitate collaboration.

7) ☐ **Data Privacy and Security**

Anonymize or pseudonymize sensitive information to protect privacy and Implement appropriate security measures to safeguard data during preprocessing and storage.

8) ☐ **Validation and Cross-Validation**

Validate data preprocessing steps using cross-validation techniques to ensure robustness and generalizability. Furthermore, Monitor for overfitting or underfitting during model training.